

REZA GHAVIPOUR

Senior Backend Developer & SRE Specialist

 Mashhad, Iran  bghavipor@gmail.com  +98 939 701 1659  linkedin.com/in/reza-ghavipour
 rghavipour.github.io/cv

• AVAILABLE FOR REMOTE & CONTRACTS

EXECUTIVE SUMMARY

Results-driven Backend Developer with **3+ years of production experience** architecting, scaling, and maintaining mission-critical microservices, high-concurrency RESTful APIs, and distributed systems using **Node.js** and **NestJS**. Demonstrated track record in successfully refactoring monolithic architectures into event-driven microservices via **RabbitMQ**, implementing sub-millisecond multi-tier caching with **Redis**, and optimizing complex relational database schemas in **PostgreSQL**. Experienced in deploying real-time computer vision AI pipelines (including inference optimization of **SAM2** and **SAM3** models) and applying **Site Reliability Engineering (SRE)** practices with **Docker** and **Nginx** to guarantee 99.9% system availability, sub-second latency, and horizontal scalability.

CORE TECHNICAL COMPETENCIES

Languages & Frameworks: TypeScript, JavaScript (Node.js), NestJS, Express.js, PHP, Laravel, Bash Scripting
Databases & Caching: PostgreSQL, MySQL, Redis (In-Memory Caching, Pub/Sub, State Synchronization), NoSQL
Architecture & Distributed: Microservices Architecture, Event-Driven Design, RabbitMQ, WebSockets, RESTful APIs, Monorepos
DevOps, Cloud & SRE: Docker, Docker Compose, Linux (LPIC-1), Nginx (Reverse Proxy/SSL), AWS (S3, SES), CI/CD
Engineering Standards: Clean Code, SOLID Principles, Design Patterns, TDD, Unit & Integration Testing (Jest), OpenAPI, ADRs
Specialized Integrations: Prisma ORM, TypeORM, Agora Video SDK, Stripe & PayPal Gateways, SAM2 & SAM3 AI Models

WORK EXPERIENCE

Backend Developer | [Manshoor Fanavari \(IFTPRISM\)](#) Jul 2022 – Present | Mashhad, Iran

- Spearheaded the refactoring of a core monolithic backend into resilient microservices with **NestJS** and **RabbitMQ**, eliminating critical deployment bottlenecks and boosting system horizontal scalability.
- Architected and deployed a real-time communication platform integrating video/audio SDKs supporting high concurrent user sessions with ultra-low latency.
- Engineered, normalized, and tuned production **PostgreSQL** databases, designing optimized indexing strategies that reduced execution times on complex reporting queries by over 45%.
- Implemented a distributed caching and session synchronization layer with **Redis**, offloading 40%+ read traffic from primary database clusters during peak load.
- Engineered real-time object tracking agents utilizing computer vision models, including performance optimization and inference tuning of **SAM2** and **SAM3** for live video processing pipelines.
- Automated containerized deployment workflows with **Docker**, **Nginx** reverse proxying, automated daily off-site backups, and SRE resilience monitoring.
- Championed Agile best practices and Git Flow within engineering sprints; authored comprehensive OpenAPI specs and Architecture Decision Records (ADRs).

Backend Developer | [Vakilnovin](#) Nov 2020 – Jul 2022 | Mashhad, Iran

- Designed and implemented the core backend for an online legal marketplace, including client dispute workflows, escrow transactions, and Telegram bot intake automations.
- Conducted query profiling and eliminated N+1 query bottlenecks via eager loading and query caching, preserving 99.8% platform uptime.
- Authored automated unit and integration test suites using **Jest** to safeguard payment processing endpoints against regressions.
- Maintained continuous delivery pipelines and contributed to structured technical documentation for developer onboarding.

SPECIALIZED CLIENT & PROJECT CAPABILITIES

- ✓ **High-Performance APIs:** Scalable REST & WebSocket microservices (NestJS, Node.js)
- ✓ **Payment Solutions:** Stripe, PayPal, Escrow billing & webhook reconciliation
- ✓ **Real-Time Video/Audio:** Low-latency streaming integration with Agora SDK
- ✓ **System Migration:** Monolith-to-microservices decomposition with RabbitMQ
- ✓ **Database Architecture:** PostgreSQL schema design, indexing & query optimization
- ✓ **DevOps & Infrastructure:** Docker, Nginx reverse proxy, CI/CD & SRE resilience

KEY PRODUCTION PROJECTS

Rose Physio Hub — All-in-One Healthcare Practice Management Platform

rosephysiohub.com

Tech Stack: NestJS Monorepo, WebSockets, RabbitMQ, PostgreSQL, Amazon S3, Stripe & Webhooks, AWS SES, Docker

- Architected a scalable backend utilizing a **NestJS monorepo** to unify practice management, appointment scheduling, and electronic records across multiple clinics.
- Engineered real-time chat modules using **WebSockets** for encrypted, low-latency communication between physiotherapists and patients.
- Built an asynchronous background job processing pipeline using message queues to handle heavy PDF medical report exports and automated patient notifications via **AWS SES**.
- Integrated **Stripe and Stripe Webhooks** for automated recurring subscription billing, dynamic invoicing, and secure multi-currency transactions.
- Secured patient records and sensitive document storage using HIPAA-ready encrypted storage on **Amazon S3**.
- Configured automated **CI/CD pipelines** with multi-stage **Docker** builds, ensuring zero-downtime rolling production deployments.

Proco — Global Mentorship & Video Consultation Platform

procoapp.com

Tech Stack: NestJS, Prisma ORM, PostgreSQL, Agora Video SDK, PayPal SDK, Redis, Docker

- Developed end-to-end backend microservices for an international mentorship platform (similar to ADPList) matching global mentors with mentees.
- Integrated the **Agora Video SDK** to provide ultra-low-latency 1-on-1 and group video consultation calls with automated token generation and access controls.
- Implemented global payment flows and automated mentor payout scheduling utilizing the **PayPal SDK**.
- Constructed strict type-safe database schemas, migrations, and complex relation queries with **Prisma ORM** and PostgreSQL.

Leaner — Agile Project Management Suite

Lead Backend Developer

Tech Stack: Node.js, NestJS, PostgreSQL, Redis, RESTful API

- Lead developer on backend services, streamlining cross-functional task workflows, sprint planning, and estimation algorithms.
- Implemented strict Role-Based Access Control (RBAC) and integrated modular internal documentation management engines.
- Boosted API response speeds by 35% through query refactoring, eager loading, and in-memory Redis caching.

VERIFIED CERTIFICATIONS

Task-Oriented Course In Linux LPIC-1

Quera • Issued May 2026 | ID: **w5Qv1fS**

Linux System Administration, Bash, SRE • [Verify Credential](#)

Associate DevOps Engineer

Arvan Cloud Academy • Issued Sep 2024 | ID: **6893**

Cloud Infrastructure, CI/CD, Docker • [Verify Credential](#)

Task-Oriented Course in NoSQL

Quera • Issued Nov 2024 | ID: **IisM1KaA**

Non-relational Data Modeling & Scalability • [Verify Credential](#)

EDUCATION

Associate Degree in Software Engineering

Montazeri Technical College of Mashhad

Jan 2017 – Jan 2019 | Mashhad, Iran

LANGUAGES & AVAILABILITY

English: Professional Working Proficiency

Persian: Native or Bilingual

⚡ Available for Contract & Full-Time Remote roles